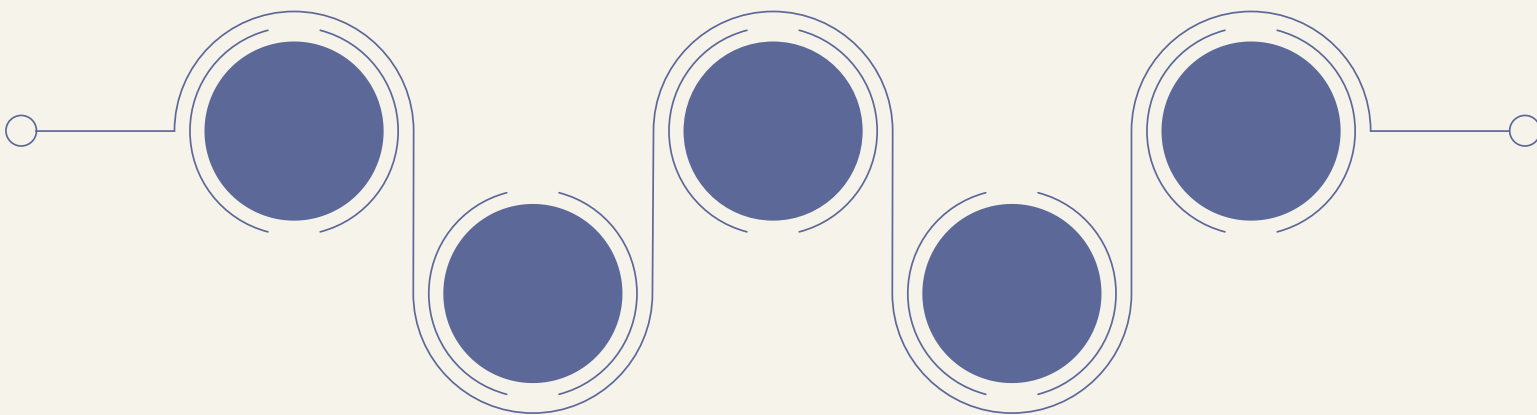




FACE RECOGNITION

Male and Female Classification

Presented by
Jasmine





Problem

Klasifikasi jenis kelamin dari citra wajah secara manual tidak efisien dan sering tidak akurat akibat variasi ekspresi, pencahayaan, dan sudut pengambilan gambar.

Goal

Mengevaluasi kinerja model deep learning VGG untuk menghasilkan klasifikasi gender yang akurat dan andal.

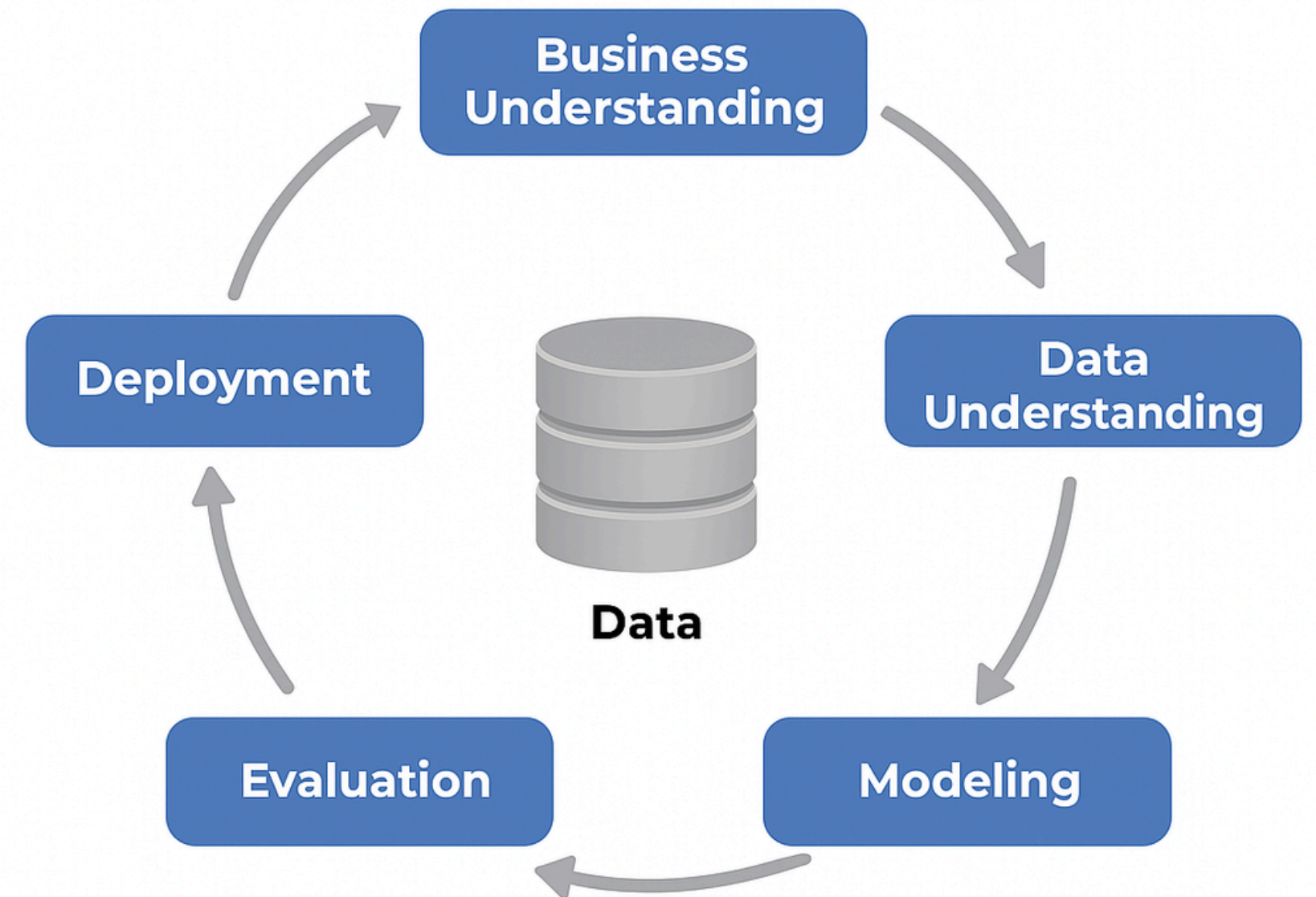
Solution

Melatih dan mengevaluasi model VGG menggunakan dataset citra wajah untuk menilai akurasi, performa, serta kelayakannya sebagai sistem klasifikasi gender otomatis.

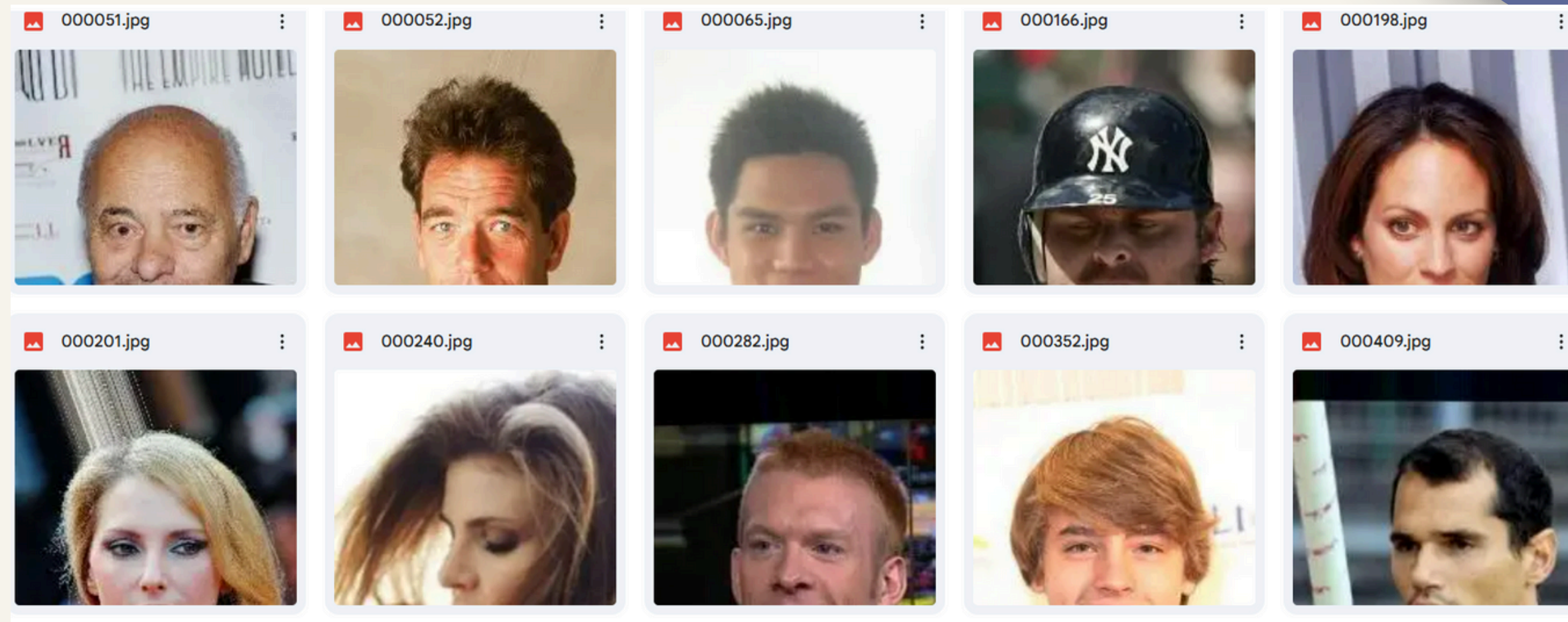


Overview

1. **Business & Data Understanding**
2. **Data Cleaning dan Analysis**
3. **Data Modeling**
4. **Model Evaluation**
5. **Model Deployment & Maintenance**

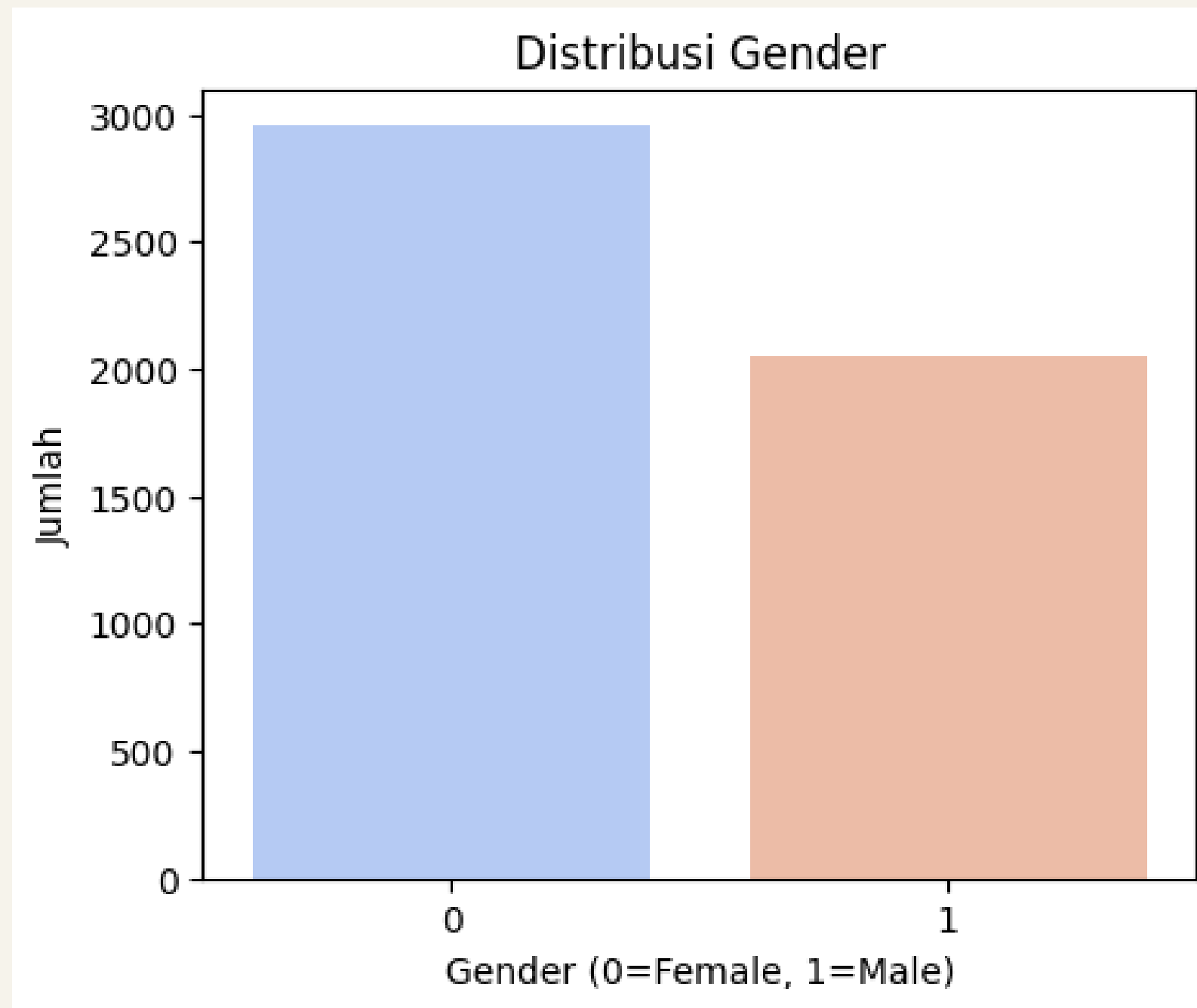


1. Business and Data Understanding



- Man and Female Classification
- Terdiri dari 5017 data gambar
- Data List Attribute : 202.601 data

1. Business and Data Understanding

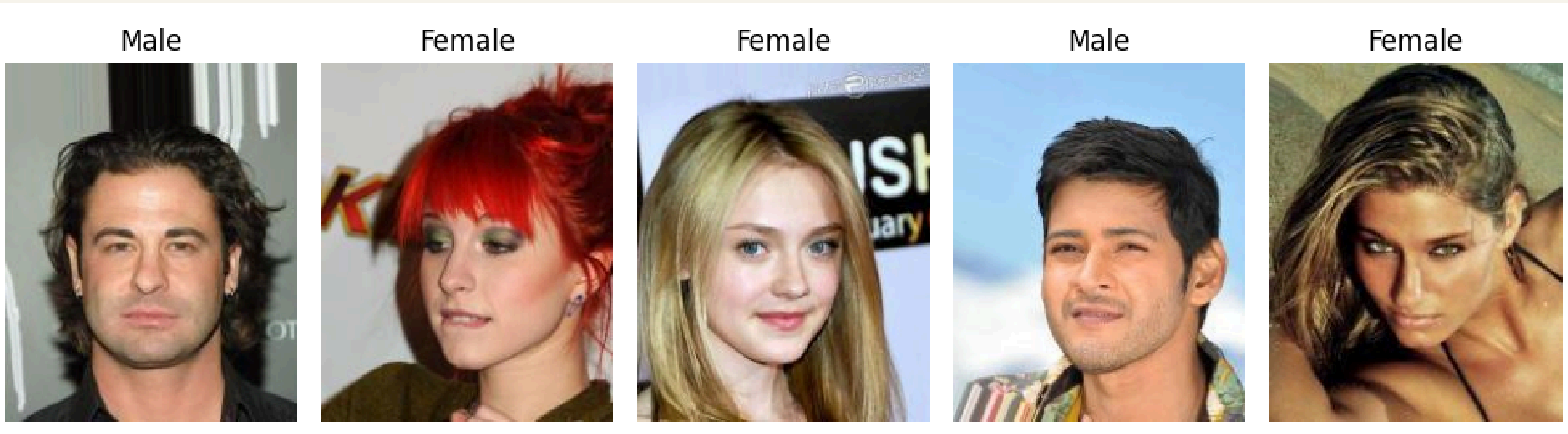


Female: 2953

Male: 2047

1. Business and Data Understanding

Contoh Gambar



2. Data Cleaning & Analysis





3. Data Modeling

VGG (Visual Geometry Group)





Result

VGG (Visual Geometry Group)

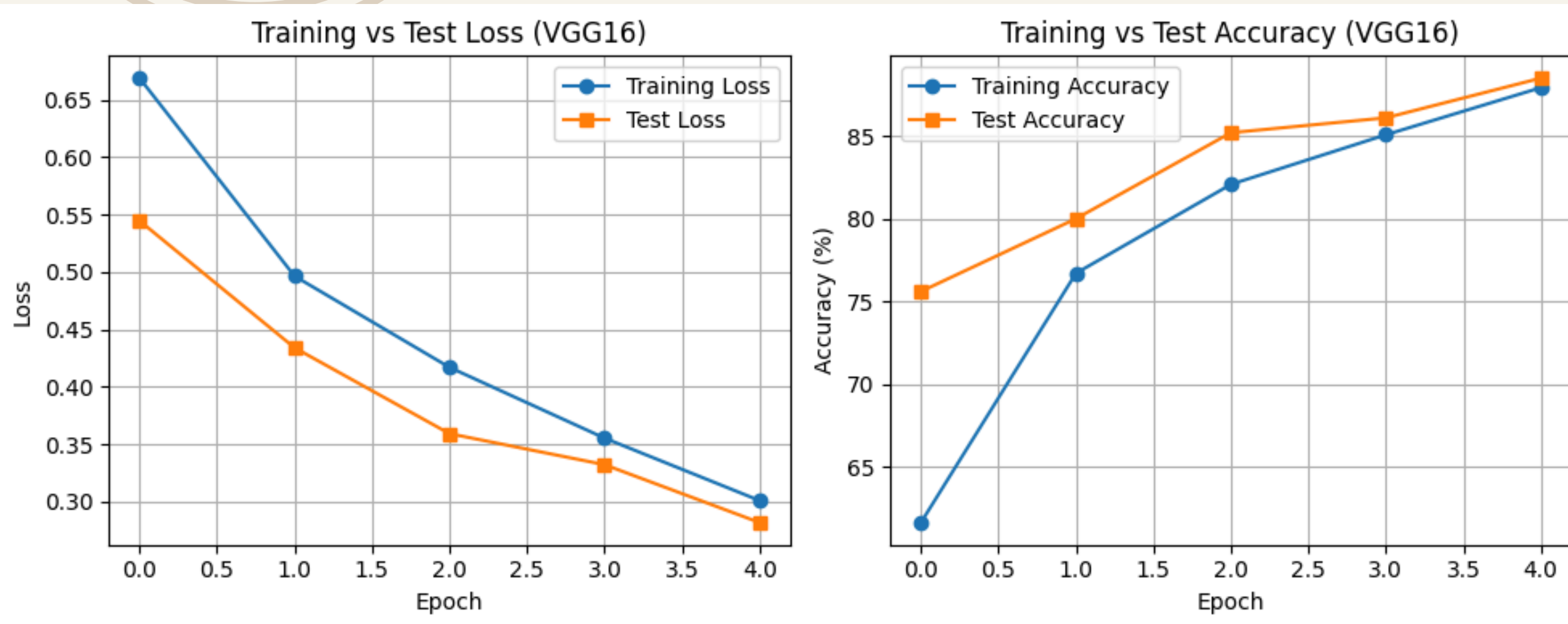
by: Jasmine

```
batch_size = 64  
learning_rate = 0.001  
epochs = 5
```

Epoch [1/5]	Train Loss: 0.6688		Test Loss: 0.5445		Train Acc: 61.62%		Test Acc: 75.60%
Epoch [2/5]	Train Loss: 0.4964		Test Loss: 0.4340		Train Acc: 76.70%		Test Acc: 80.00%
Epoch [3/5]	Train Loss: 0.4169		Test Loss: 0.3592		Train Acc: 82.08%		Test Acc: 85.20%
Epoch [4/5]	Train Loss: 0.3553		Test Loss: 0.3322		Train Acc: 85.08%		Test Acc: 86.10%
Epoch [5/5]	Train Loss: 0.3010		Test Loss: 0.2818		Train Acc: 87.92%		Test Acc: 88.50%



VGG (Visual Geometry Group)



Grafik menunjukkan model VGG16 belajar dengan baik, ditandai penurunan loss dan peningkatan akurasi tanpa overfitting.

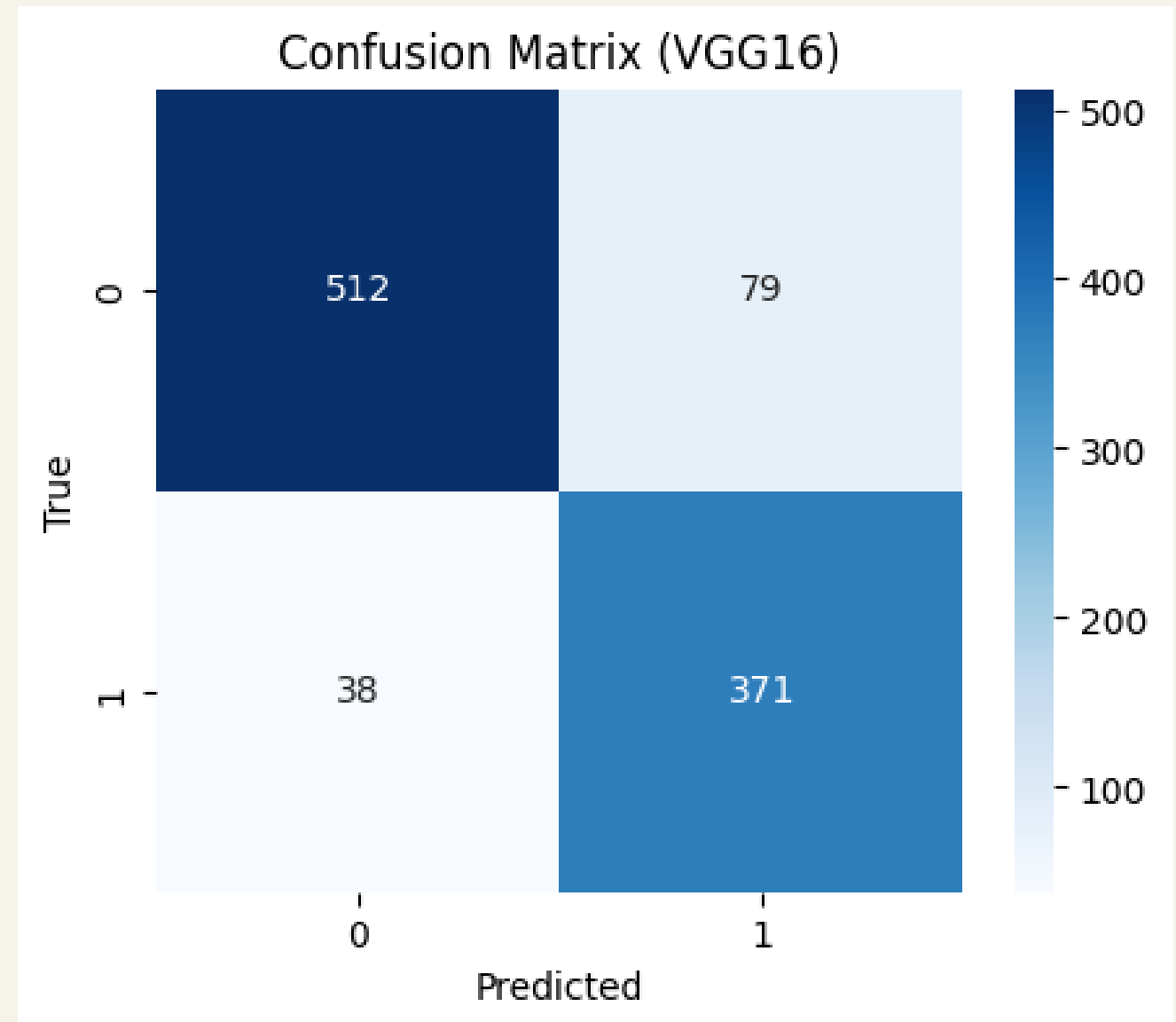
VGG (Visual Geometry Group)

=== Classification Report ===

	precision	recall	f1-score	support
0	0.93	0.87	0.90	591
1	0.82	0.91	0.86	409
accuracy			0.88	1000
macro avg	0.88	0.89	0.88	1000
weighted avg	0.89	0.88	0.88	1000

Model mencapai akurasi 88% dengan kinerja seimbang; kelas 0 lebih presisi, kelas 1 lebih baik recall.

Model VGG16 mengklasifikasikan data dengan akurat, dengan 512 benar kelas 0, 371 benar kelas 1, serta 79 dan 38 prediksi salah.





THANK YOU

